

Generation Interconnection Request System Impact Study Report

For Queue Project Q18-1079

Amelia St 100 kV
August 25, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1079. The Interconnection Customer proposes a 137.8 MW solar photovoltaic facility interconnecting at the Amelia St 100 kV substation in the Indiana area. The request was received on October 15, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 305 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$642,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Dyckesville Substation / E Jackson 1&2 100kV Line
Contingency: Outlook / Hegewisch 100kV Line
Loading: 122.5% (Rating: 109 MVA)
DFAX: 36.0%
Pre-project loading: 77.0%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Dyckesville Substation / E Jackson 1&2 100kV Line	Rebuild line with higher-capacity conductor	\$1,911,000	34% (\$642,000)
	Total allocated to project		\$642,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.